

# precog-esgf-intake: Automated discovery, validation, and optimized download management of Earth system model data from Earth System Grid Federation nodes.

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## Software

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## Summary

precog-esgf-intake is a Python command-line-interface (CLI) software for automated discovery, checking, validation, and download of Earth System Model outputs from Earth System Grid Federation (ESGF) archives. The software is designed for research workflows that require reproducible access to large, distributed climate-model datasets and is particularly aimed at bulk screening of Earth System archives before downstream analysis. The software builds on inherited ESGF access functionality from intake-esgf (Collier et al., 2026), while introducing interactive CLI workflows for archive interrogation, file-availability checks, grid-consistency cross-checks, temporal validation, export of search results in tabular formats, and optimised download management of shortlisted Earth System Model data products.

## Statement of need

Modern Earth System science workflows frequently rely on climate-model data distributed across Earth System Grid Federation (ESGF) nodes (Cimadevilla et al., 2025). For example, a Coupled Model Intercomparison Project (CMIP) analysis to constrain future projections in ocean carbon inventories requires standardized output of ocean biogeochemical variables (carbon, oxygen, nutrients) across many models and experiments (Wilson et al., 2022). These archived data are often distributed across many federated storage nodes, including duplicated, incomplete or corrupted versions. Although ESGF provides a federated infrastructure for searching and accessing these archives, practical research workflows often require additional automation to determine whether a given combination of variables, experiments, ensemble members, and grid configurations is both scientifically suitable and operationally downloadable before substantial time is spent retrieving files.

This issue is especially important for CMIP-style analyses in which researchers may need to confirm that the same Earth System Model provides paired pre-industrial, historical and ssp-scenario simulations, that multiple variables of interest are available and whether these are separately archived on compatible grids for downstream analyses; and that temporal coverage is continuous across archived files. Manual inspection of catalogue search results can become slow, repetitive, and error-prone when screening many candidate models or variables across multiple ESGF nodes.

Among its features, precog-esgf-intake provides an interactive workflow for shortlisting Earth System Model datasets that satisfy compound criteria across experiments and variables. The software validates temporal coverage, grid consistency, and file availability across published ESGF archives before download. Users can export search results as tabular summaries, inspect

41 and refine shortlisted datasets, and initiate batch downloads or trigger file integrity checks on  
42 locally assigned output paths. Retrieved data are then organised into a directory structure  
43 suitable for reproducible downstream analysis. This workflow is particularly useful when  
44 researchers need to screen many candidate models and variables before selecting datasets that  
45 are both scientifically appropriate and operationally accessible within their computational and  
46 storage constraints.

47 Furthermore, by automating the retrieval and validation of CMIP grid metrics like `volcello`  
48 , `areacello` as default (and enabling further bespoke searches for `deptho` or `thkcello` as  
49 a validated substitutes when `volcello` is unavailable) — `precog-esgf-intake` reduces the  
50 need for users to understand CMIP metadata and ESGF directory conventions. By wrapping  
51 ESGF discovery in a simple Python CLI and allowing users to tailor searches by editing a  
52 straightforward `search_criteria.toml` configuration file, the package supports both accessible  
53 default workflows and more bespoke dataset selection. This lowers the barrier for researchers  
54 new to CMIP while enabling experienced users to specify models, experiments, variables,  
55 frequencies, and other search constraints. Its lightweight design also supports terminal-based  
56 HPC workflows, allowing data discovery and ingestion from visualisation nodes and Jupyter  
57 sessions rather than interactive web portals or manual downloads.

## 58 State of the field

59 CMIP data are currently accessible through the ESGF metagrid web application, which provides  
60 web-based discovery and download services for CMIP5, CMIP6, CMIP6Plus, and the upcoming  
61 CMIP7 phases. Recent development efforts have focused on analysis-ready, cloud-optimised  
62 approaches based on in-memory object storage (Mizielinski et al., 2026), with catalogue  
63 sweep tools enabling more scalable access through Python and `xarray`, while community  
64 evaluation frameworks like `ESMValTool` (Righi et al., 2020) supporting integrated diagnostics  
65 for benchmarking model outputs.

66 `precog-esgf-intake` builds on the ESGF catalogue node sweeping implementation from  
67 `intake-esgf` (Collier et al., 2026), but targets a different level of the workflow: rather than  
68 focusing only on data access primitives and caching data-responses to in-memory use, it  
69 provides an interactive command-line environment for ample discovery, screening, validating  
70 and managing downloads of published ESGF archive products that satisfy scientific constraints  
71 relevant to Earth system and ocean biogeochemistry applications. The software is intended for  
72 situations in which researchers need to move efficiently from broad discovery and catalogue  
73 searches to a smaller, analysis-ready subset of model output that satisfies practical and scientific  
74 constraints.

## 75 Key functionality

76 The software provides several features tailored to archive-scale Earth system data workflows:

- 77 ■ Definition of ESGF search criteria by modifying a `search_criteria.toml` configuration  
78 file.
- 79 ■ Automated search through the ESGF catalogue using project, variable, experiment, and  
80 temporal filters (inherited from `intake-esgf`).
- 81 ■ Ensemble-aware screening mode for selecting one validated `piControl` anchor branch  
82 and retaining all compatible `Historical` ensemble members on the same grid.
- 83 ■ Single-branch screening mode for selecting one internally consistent `piControl`/`Historical`  
84 branch pairing per accepted model/grid.
- 85 ■ Combined-variable screening for retaining models that provide all requested variables  
86 simultaneously across both `piControl` and `historical` runs, producing a single shortlist  
87 for downstream download.

- 88     ▪ Verification and logging of continuity of date stamps and availability of piControl and
- 89     Historical runs on consistent grids (e.g., regular grid gr and native grid gn).
- 90     ▪ Export of simple Dataframes for realised ESGF catalogue searches.
- 91     ▪ Verification of server responses and flagging shortlisted ESM outputs as 'Downloadable'.
- 92     ▪ Parallelized URL checks for fastest connection in case same data are available across
- 93     different nodes.
- 94     ▪ Parallel batch downloading of files from multiple ESGF nodes with retry and integrity
- 95     checks.
- 96     ▪ Parallel batch search and download of grid cell measures (e.g., areacello and volcello)
- 97     with archive snapshot of relaxed regex matches for later inspection.
- 98     ▪ Local directory layout optimised for downstream analysis.
- 99     ▪ Example Jupyter notebook workflow covering TOML setup, catalogue screening, and
- 100    download preparation.

## 101   Software design and workflow overview

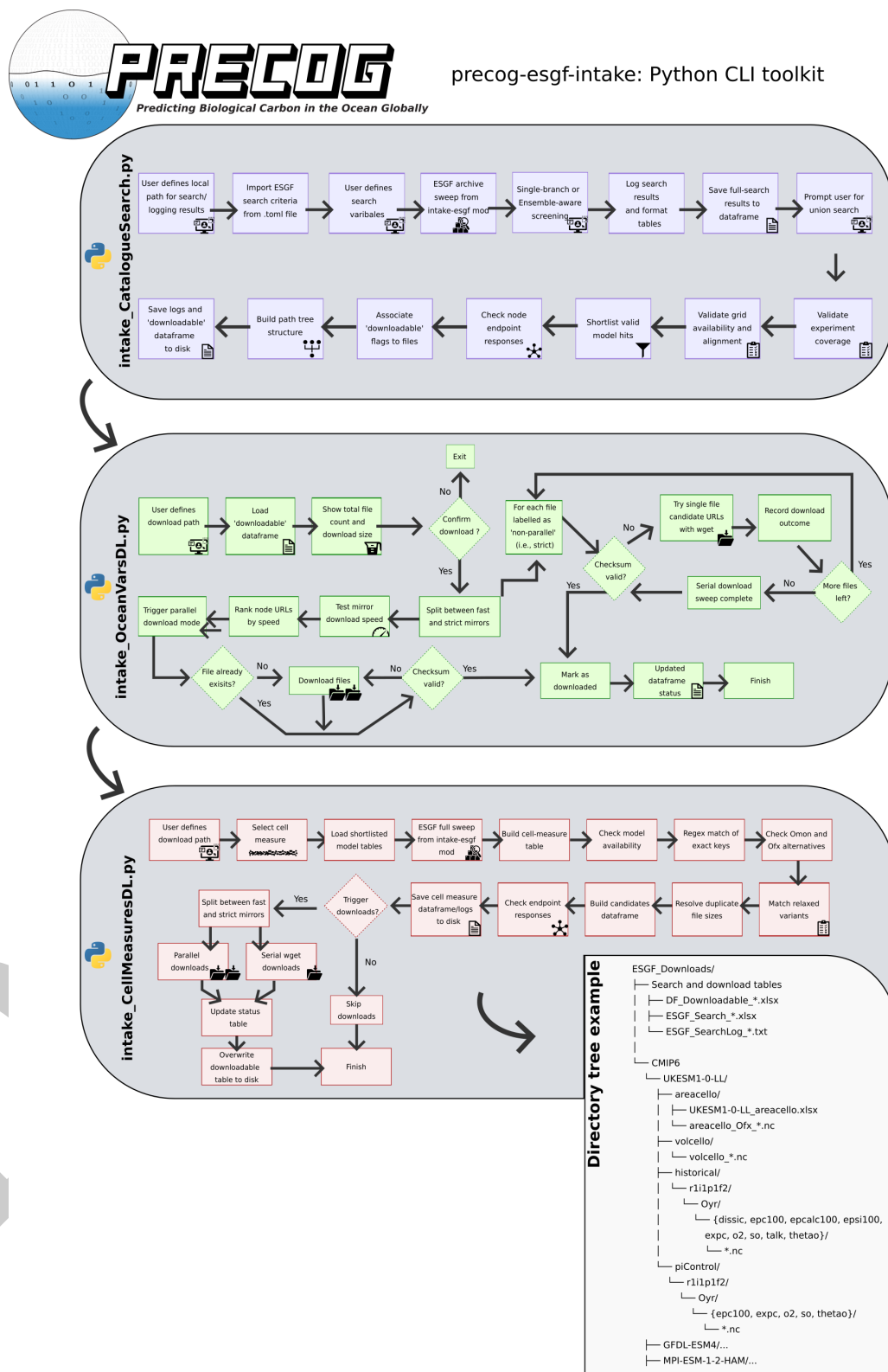
102   precog-esgf-intake implements a staged workflow for archive-scale ESGF data discovery,  
 103   screening, and retrieval. Rather than moving directly from catalogue search to cached download,  
 104   the software separates archive interrogation, shortlist generation, branch and grid validation,  
 105   downloadability checks, and file retrieval into distinct command-line steps, allowing users to  
 106   inspect and validate intermediate results before proceeding. This design follows common  
 107   scientific-workflow principles in which complex analyses are decomposed into explicit tasks  
 108   with traceable intermediate outputs, improving transparency, reuse, and reproducibility.

109   In a typical workflow (Figure 1), the user first edits a user-facing `search_criteria.toml`  
 110   configuration file to define the intended ESGF search criteria, including project, activity,  
 111   experiment, frequency, variables (optional), and grid labels. This externalised configuration  
 112   makes search intent easier to reproduce than editing hard-coded dictionaries in source code,  
 113   while still allowing interactive entry of `variable_id` when that field is intentionally left blank.  
 114   The catalogue search stage then queries ESGF holdings for matching CMIP products (CMIP6 by  
 115   default), exports tabular search summaries for inspection, and applies screening logic to retain  
 116   scientifically relevant combinations such as paired piControl and historical simulations on  
 117   common valid grids.

118   A key feature of the workflow is the ability to **combine** multiple requested variables into a single  
 119   conditional shortlist. Rather than screening each variable independently, users can require  
 120   that the same model provides all requested variables simultaneously, for both piControl  
 121   and historical, before that model progresses to the downloadable shortlist. This combined  
 122   screening is particularly useful for multi-field diagnostics where consistent co-availability of  
 123   variables is required for downstream interpretation.

124   The workflow supports both single-branch and ensemble-aware catalogue screening. In single-  
 125   branch mode, the search retains one internally consistent piControl/historical branch  
 126   pairing for each accepted model-grid combination. In ensemble-aware mode, the workflow first  
 127   identifies a validated piControl anchor branch and then retains all compatible historical  
 128   ensemble members on the same model/grid pair, provided that the requested variables are  
 129   complete and continuity checks are satisfied. Grid-level diagnostics are generated explicitly from  
 130   catalogue metadata so that invalid model-grid combinations can be rejected before download  
 131   preparation, and validation outputs are preserved as tabular artefacts for later auditing.

132   Subsequent stages verify whether shortlisted files are reachable on remote ESGF nodes, assign  
 133   local destination paths, and download both target variables and required supporting grid-cell  
 134   measures such as areacello and volcello. By treating search results, validation outputs, and  
 135   downloadable file lists as explicit intermediate artefacts, the software supports both interactive  
 136   exploratory use and more reproducible archive-based Earth system workflows in which dataset  
 137   selection decisions can be inspected, traced, and repeated at a later time.



*Figure 1. precog-esgf-intake toolkit overview and directory structure of an example ESGF download. The top-level directory contains search outputs and model-specific CMIP6 data organised by model, experiment, variable, and annual files.*

## Research impact and applications

A representative use case is the identification of CMIP6 models that simultaneously provide both pre-industrial and historical outputs for ocean biogeochemical variables such as `expc` and `epc100`, together with auxiliary or supporting variables and the associated grid-cell measures required for downstream analyses. The repository accompanying this software includes a [workflow example](#) demonstrating this type of archive screening and retrieval process.

This is particularly relevant for ocean biogeochemistry and carbon-cycle studies, where analyses often depend on coherent combinations of physical and biogeochemical fields rather than isolated variables (e.g., retrieval of carbonate system fields as well as ocean state physical variables, (Wilson et al., 2022)). In such cases, the time spent screening archive holdings, checking consistency, and organising downloads can be substantial, and purpose-built automation improves both efficiency and reproducibility. This data discovery and retrieval functionality by `precog-esgf-intake` is also complementary to established CMIP preprocessing packages such as `xMIP` (Busecke et al., 2023) and `ESMValTool` (Righi et al., 2020), which provide tools for harmonising, cataloguing, and preparing model outputs for analysis; together, these packages support a reproducible workflow from archive discovery and dataset selection through preprocessing and scientific analysis.

Therefore, `precog-esgf-intake` fills a workflow gap between catalogue access and scientific analysis. The software capabilities support workflows in which data access is itself a significant part of the scientific process, particularly when analyses depend on assembling coherent, scientifically consistent, and analysis-ready ensemble subsets of Earth system model output from distributed archives.

## AI usage disclosure

AI assistance was used solely to improve readability of selected software documentation (Claude Sonnet 5 by Anthropic). No AI tools were used to design, implement, or validate the software. All AI-assisted documentation was reviewed and edited by the authors, who retain full responsibility for its accuracy and content. Authors made all the core design and architectural decisions.

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